

CASEY SKIWAY SOUTH, Antarctica

WMO: 898090

Lat: 66.285S

Lon: 110.743E

Elev: 1230

StdP: 14.05

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-29.2	-24.3	N/A	N/A	N/A	N/A	N/A	N/A	73.0	9.2	67.2	11.7	8.2	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	37.9	N/A	35.0	N/A	33.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 65.3	56.3	48.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	26.7	21.7	12.0	7.1	1.1	2.0	2.2	-0.7	3.2	6.1	16.0	24.1
	DBStd	12.93	4.61	5.23	8.49	8.47	11.20	11.13	11.26	9.98	10.31	8.73	7.74	6.62
	HDD50	14581	721	792	1178	1286	1516	1441	1483	1573	1405	1362	1020	804
	HDD65	20056	1186	1212	1643	1736	1981	1891	1948	2038	1855	1827	1470	1269
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.6	12.7	13.9	13.7	14.3	15.7	18.0	26.2	17.7	18.6	17.2	15.5	14.7
Precipitation	PrecAvg	18.0	1.5	1.6	1.6	1.4	1.3	1.5	1.8	1.8	1.4	1.2	1.5	1.5
	PrecMax	23.1	2.2	2.6	2.8	2.8	2.1	2.7	3.1	3.0	2.4	2.0	2.8	2.8
	PrecMin	14.3	0.8	0.2	0.7	0.5	0.3	0.5	0.8	0.7	0.6	0.4	0.4	0.6
	PrecStd	2.2	0.4	0.6	0.5	0.6	0.4	0.5	0.7	0.6	0.4	0.4	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	38.8	28.9	25.6	24.9	23.3	25.3	24.4	29.7	26.3	35.0	42.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	38.2	33.3	26.1	23.1	21.2	21.1	21.6	18.8	21.6	22.4	31.0	37.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	35.6	31.3	24.3	20.9	19.4	19.4	18.8	14.6	18.2	19.6	28.1	34.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	33.7	29.5	22.7	18.7	16.6	16.9	16.2	12.1	16.2	17.6	25.8	32.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	13.8	13.1	14.0	12.3	12.9	12.7	12.2	12.4	12.6	16.7	16.1	15.4
		MCDBR	17.2	17.4	13.1	11.4	11.9	11.9	10.2	13.4	9.3	16.7	20.1	21.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.253	0.243	0.222	0.184	0.122	0.122	0.122	0.126	0.181	0.210	0.219	0.238	0.254
	taud	2.342	2.380	2.360	2.221	1.926	1.926	1.926	1.901	2.169	2.275	2.293	2.298	2.289
	Ebn at Noon	321	308	284	231	123	24	106	225	283	315	325	325	
	Edh at Noon	34	29	22	13	4	1	4	14	23	31	35	37	
All-Sky Solar Radiation	RadAvg	2444	1695	892	306	53	3	22	183	669	1500	2337	2741	
	RadStd	108	93	48	23	3	0	1	12	30	44	83	93	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page